Supplementary Information (Online Resource 4)

Article title: How Teams Decide on Human Oversight in AI-Powered Systems: A Mixed-Methods Study

Journal: Empirical Software Engineering

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This resource presents the ontology-guided research interview protocol from the broader OG-RE4AI study. Its questions were used to elicit participants' experiences and perspectives concerning AI-system requirements. For the present study, oversight-related interview material was selected for focused analysis, and the protocol provided an analytical lens for examining relationships and tensions between requirements. It is not presented as a practitioner questionnaire developed or validated for use by project teams in Phase 4 of RE4HOAI.

OG-RE4AI research interview protocol

Questions shown in bold italic font identify parts of the OG-RE4AI research interview protocol relevant to the negotiation of human-oversight requirements. These questions address stakeholder roles and priorities, human involvement, explainability, trust, competing requirements, trade-offs, and the incorporation of stakeholder feedback. The marking makes their relevance to the present study explicit, while the remaining questions are retained to preserve the broader context of the research protocol.

Question group	Requirement category	Interview questions
1	General	1.1 - What was the business need or motivation that caused the initiation of the AI-driven project? 1.2 - What are the factors that influence the decision to use AI over conventional methods?
2	General	2.1 – What are the roles and responsibilities of different stakeholders in the process of developing requirements of AI systems, both FRs and NFRs? 2.2 – How can we ensure and encourage their effective participation and collaboration? 2.3 – What are the effective methods, procedures and tools for elicitation the complete requirements for AI-powered systems?
3	General	***3.1 – What are the challenges and best practices for communicating and negotiating the requirements of AI systems with different stakeholders, with a particular emphasis on user engagement?*** 3.2 – How can we ensure that their feedback is properly considered and integrated into the final product? 3.3 – How can we surmount the diverse attitudes, perceptions, and understanding of AI among stakeholders?
4	General	4.1 – How can we assess the readiness and feasibility of AI solutions for specific tasks and contexts? 4.2 – What are the initial stages of assessing the system's readiness for AI integration?
5	General	***5.1 - What are the tools and techniques that can be used for verifying and validating the requirements of AI systems?*** ***5.2 - In what stages along the system life-cycle do you validate the performances and how?*** ***5.3 – What constitutes the motivation for this validation?*** 5.4 - How can we measure and validate the ethical requirements of AI systems?
6	General	What are the primary causes of requirement changes in AI-powered systems, and what are the best practices for managing these changes?
7	General	7.1 – What are the interdependencies and contradictions between different NFRs and NFRs against FRs? ***7.2 – What are the common trade-offs between different requirements?***

		7.3 – How are they prioritized, balanced, resolved and managed?
8	General	8.1 - When should different types of requirements be defined within the project timeline? ***8.2 - What are the critical requirements to be addressed at the project initiation?***
9	General	***What are the testing methods specifically required for AI-powered systems?***
10	General	Does the organization possess the requisite resources to successfully integrate and deploy the AI solution?
11	Model	11.1 - What are the challenges and best practices for selection and managing the requirements of AI models? 11.2 –how does data quality impact model selection? 11.3 –What is the projected evolution of the model, and how can this evolution be managed strategically?
12	Model	***How can we verify that AI model deviations or failures of the system are properly detected and addressed?***
13	Data	13.1 - What are the most critical data qualities, and what data preparation is required in order to successfully implement AI capabilities? 13.2 –What potential risks are associated with the selection of specific features in the data?
14	Data	14.1 - What are the requirements for measuring and evaluating data quality and data drifts along the system life-cycle? 14.2 –What are the appropriate timing and triggers for conducting this evaluation? 14.3 –Is there a requirement for automation evaluation or is manual evaluation sufficient? 14.4 –How does noisy data impact system performance, and what strategies can be implemented to mitigate its effects?
15	Data, Model	What data and model requirements are necessary to prevent biases that could affect the system's fairness?
16	Data	16.1 – What are the roles and responsibilities of data personnel in the requirement development process for AI systems? 16.2 –How data scientists are involved in the agile project approach?
17	Performance	17.1 - What methods and techniques are employed for defining performance requirements at the project's initial phase? 17.2 –What are the categories of performance requirements and how are performance values addressed? 17.3 - How do stakeholders overcome the challenge of defining performance expectations without the ability to observe the performance until the conclusion of the training period?
18	Explainability	***18.1 - How do you define the level of explanation required for different stakeholders based on their roles in the development stages and the system operation?*** ***18.2 –What are the requisite levels of explainability for diverse stakeholders, particularly users and developers?***
19	Explainability	***19.1 What methods are used for explainability?*** ***19.2 How does explainability influence the model selection?***
20	Explainability	***20.1 - Could the explainability requirement issue be a "show stopper" for the decision not to use AI for a specific task?*** ***20.2 – What other solutions are suggested when explainability is not feasible?***
21	Ethics	21.1 - How can we ensure that ethical considerations are effectively integrated into the development and deployment of AI systems? ***21.2 - How can we ensure that ethical requirements are managed along the system life-cycle?*** 21.3 –Who needs to be involved in ethical requirements management? 21.4 –At what stage in the project lifecycle should ethical requirements be developed?
22	Ethics, Responsibility	***What measures can be taken to address violations of ethics within the system, and what is the level of accountability held by the development team and the user?***
23	Trust	***23.1 – What are the key factors that contribute to user trust in AI systems?*** ***23.2 – How do these factors vary across different contexts and applications?***
24	Trust	***What are the best practices for evaluating and measuring user trust in AI systems?***
25	Privacy	25.1 - How can existing privacy frameworks and regulations, such as GDPR, be adapted and extended to effectively govern the use of AI systems and protect privacy? 25.2 –What is the impact of including privacy requirements on system design and performance?
26	Trainability	***26.1 - What are the factors that should be considered when defining the frequency of retraining an AI-powered system?*** 26.2 –What are the tools and methods to perform the trainability? 26.3 –What tools and technologies are necessary to enable the model to undergo autonomous retraining
27	Human-system interaction	***27.1 - What is the anticipated level of human involvement in the system?*** ***27.2 - What are the primary justifications for human intervention within the system?***
28	Security	What specific measures can be implemented to address potential vulnerabilities and threats?
29	Risk mitigation	***What requirements are essential for effective risk mitigation within the AI system?***

30	Sustainability	30.1 What sustainability regulations apply to the system's domain? 30.2 How will the system measure and report compliance with sustainability metrics?